

Gebze Technical University

Dep. of Mechanical Engineering

Statics (ME 231)

Lecture 11:

Internal Forces

By Saeed Lotfan

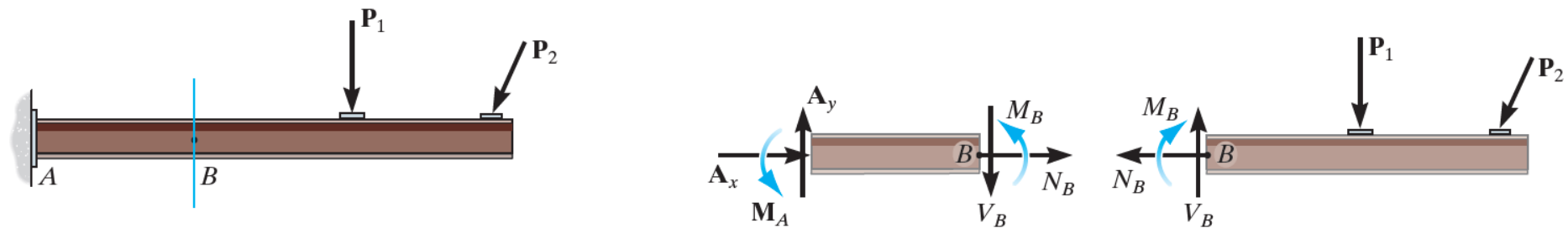
Internal Loadings



The loads within the beams and columns of this building frame must first be determined if they are to be properly designed. In this lecture we will study how to find these internal loadings.

Internal Loadings

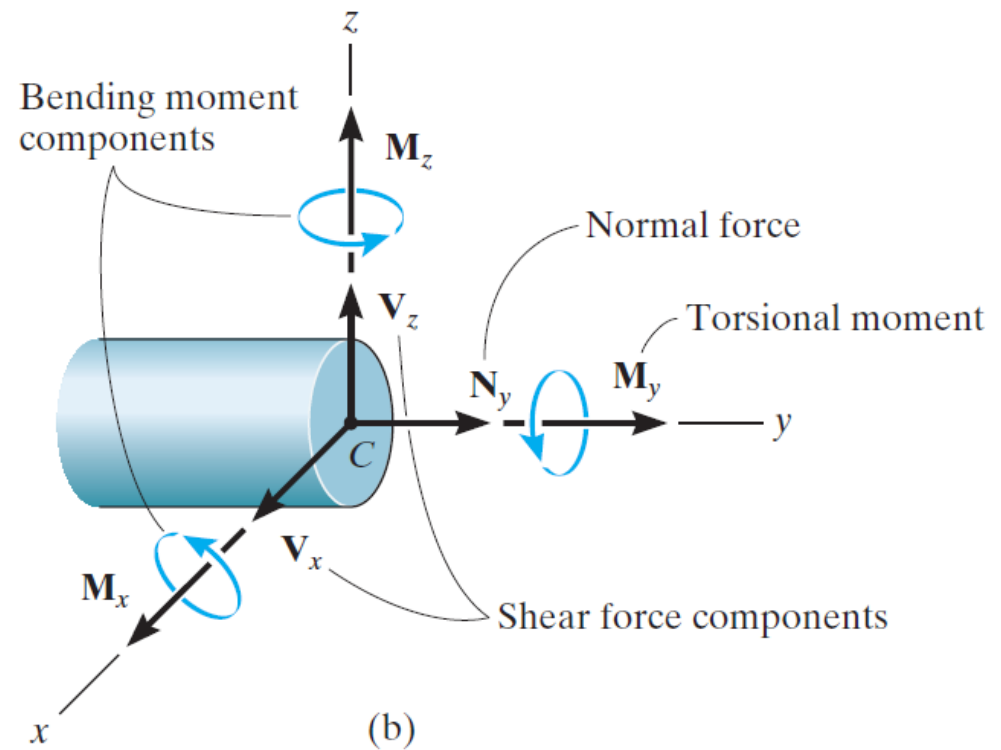
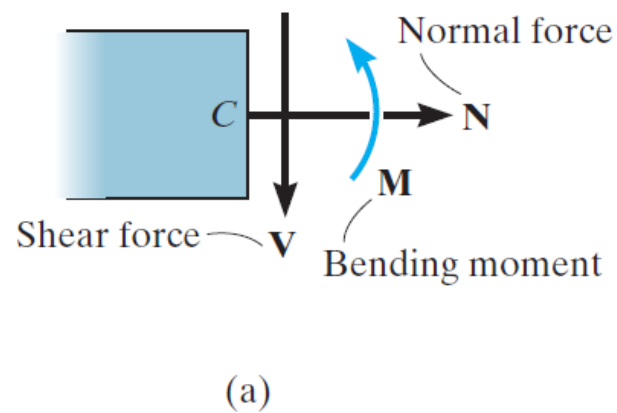
To design a structural member or mechanical element it is necessary to know the loading acting within it to be sure the material can resist this loading. Internal loadings are determined by using the ***method of sections***.



- The force component N_B that acts *perpendicular* to the cross section is termed the ***normal force***.
- The force component V_B that is tangent to the cross section is called the ***shear force***, and the couple moment M_B is referred to as the ***bending moment***.

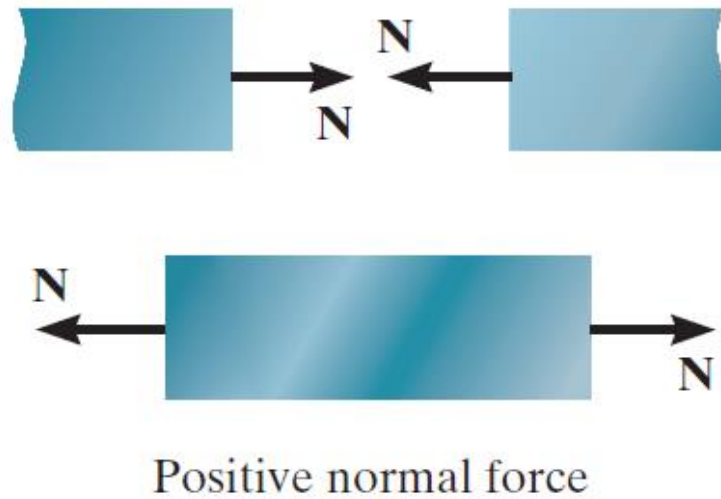
Internal Loadings

In three dimensions, a general resultant internal force and couple moment resultant will act at the section.



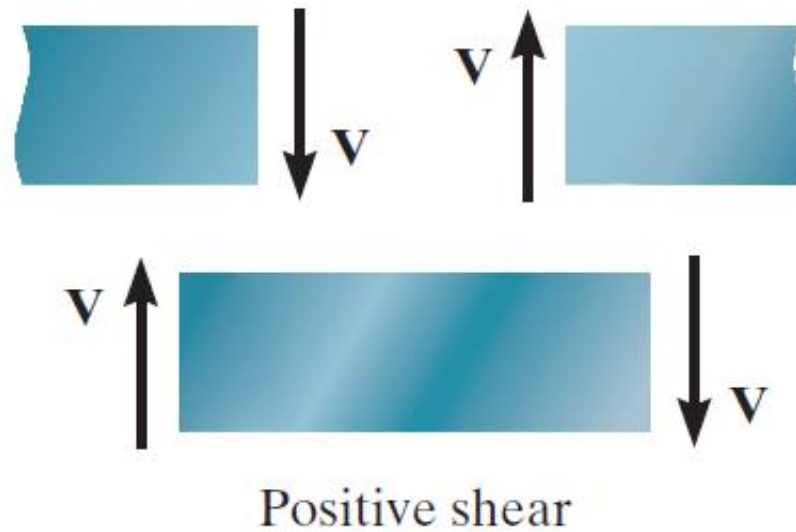
Internal Loadings

Sign Convention



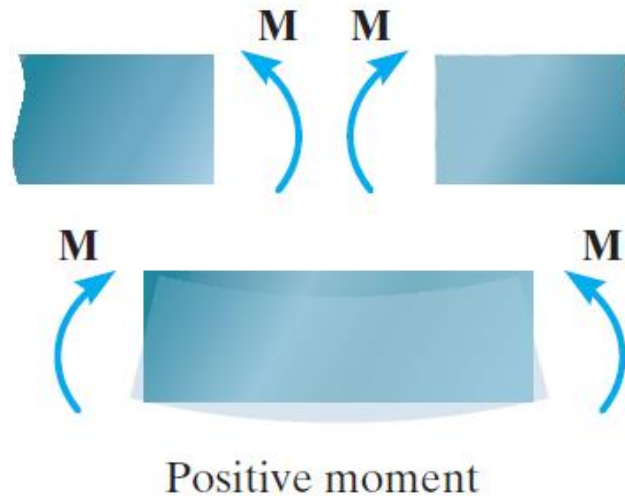
Internal Loadings

Sign Convention



Internal Loadings

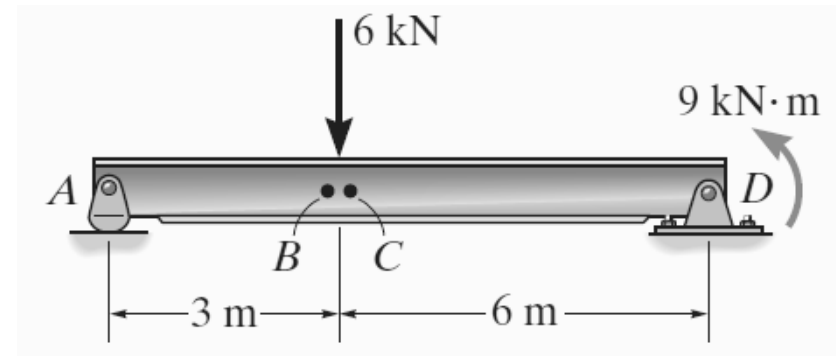
Sign Convention



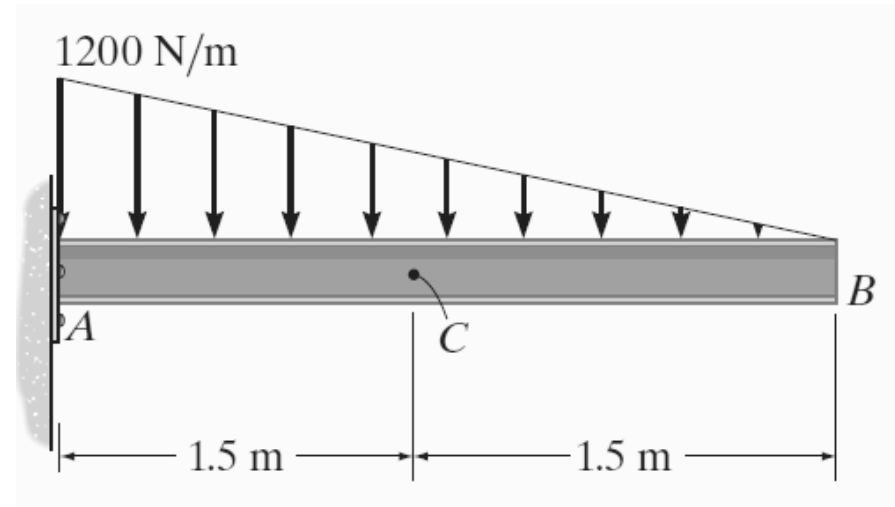
Concluding Note

- There can be four types of resultant internal loads in a member. They are the **normal** and **shear forces** and the **bending** and **torsional moments**. These loadings generally vary from point to point. They can be determined using the method of sections.

Example: Determine the normal force, shear force, and bending moment acting just to the left, point B , and just to the right, point C , of the 6-kN force on the beam shown below.



Example: Determine the normal force, shear force, and bending moment at point C of the beam below.

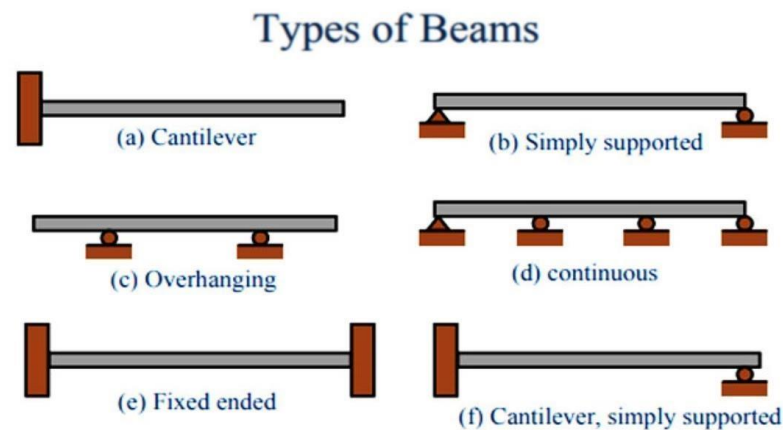


Shear and Moment Equations and Diagrams

Beams are structural members designed to support loadings applied perpendicular to their axes.

In general, they are long and straight and have a constant cross-sectional area.

TYPES OF BEAMS



The actual design of a beam requires a detailed knowledge of the *variation* of the internal **shear force** V and **bending moment** M acting at *each point* along the axis of the beam.

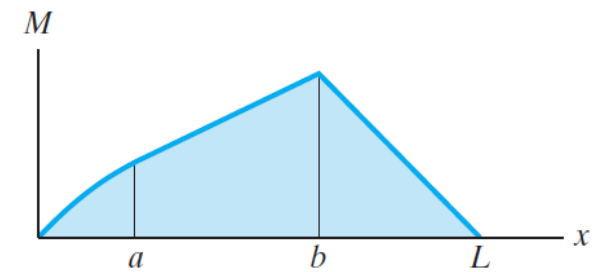
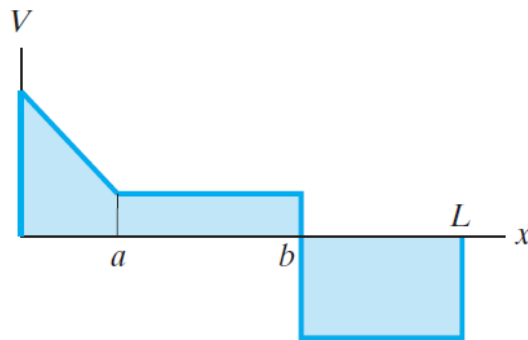
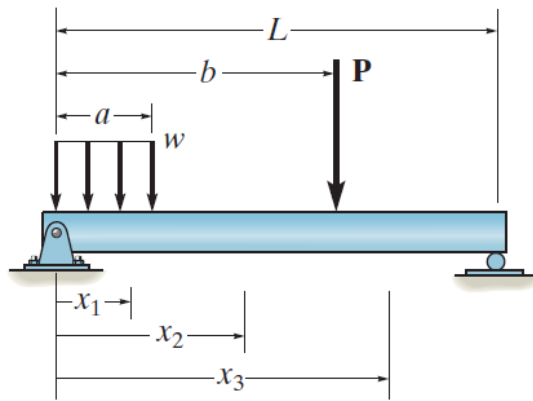


The shelving arms must be designed to resist the internal loading in the arms caused by the lumber.



To save on material and thereby produce an efficient design, these beams, also called girders, have been tapered, since the internal moment in the beam will be larger at the supports, or piers, than at the center of the span.

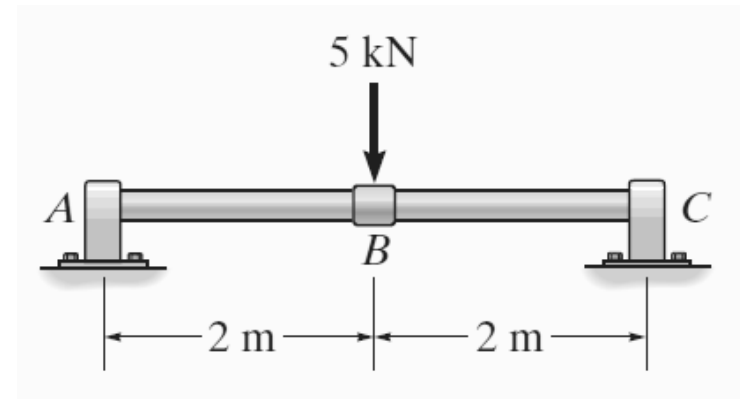
Shear and Moment Equations and Diagrams



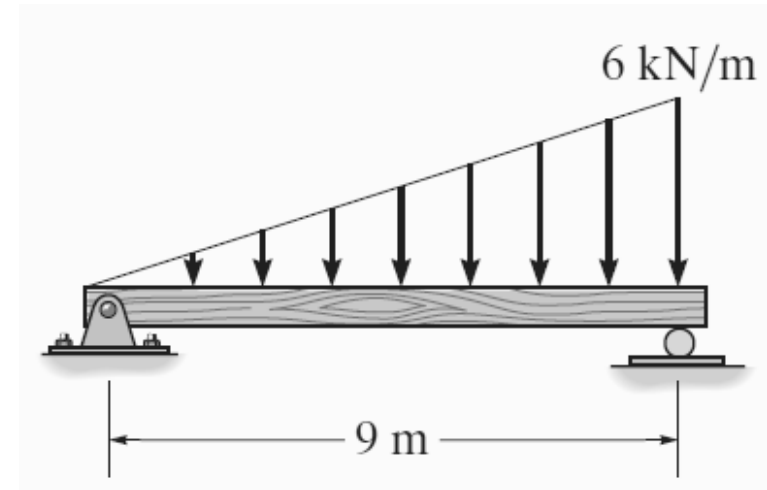
Concluding Notes

- Shear and moment diagrams for a beam provide graphical descriptions of how the internal shear and moment vary throughout the beam's length.
- To obtain these diagrams, the method of sections is used to determine V and M as functions of x . These results are then plotted. If the load on the beam suddenly changes, then regions between each load must be selected to obtain each function of x .

Example: Draw the shear and moment diagrams for the shaft shown below. The support at A is a thrust bearing and the support at C is a journal bearing.

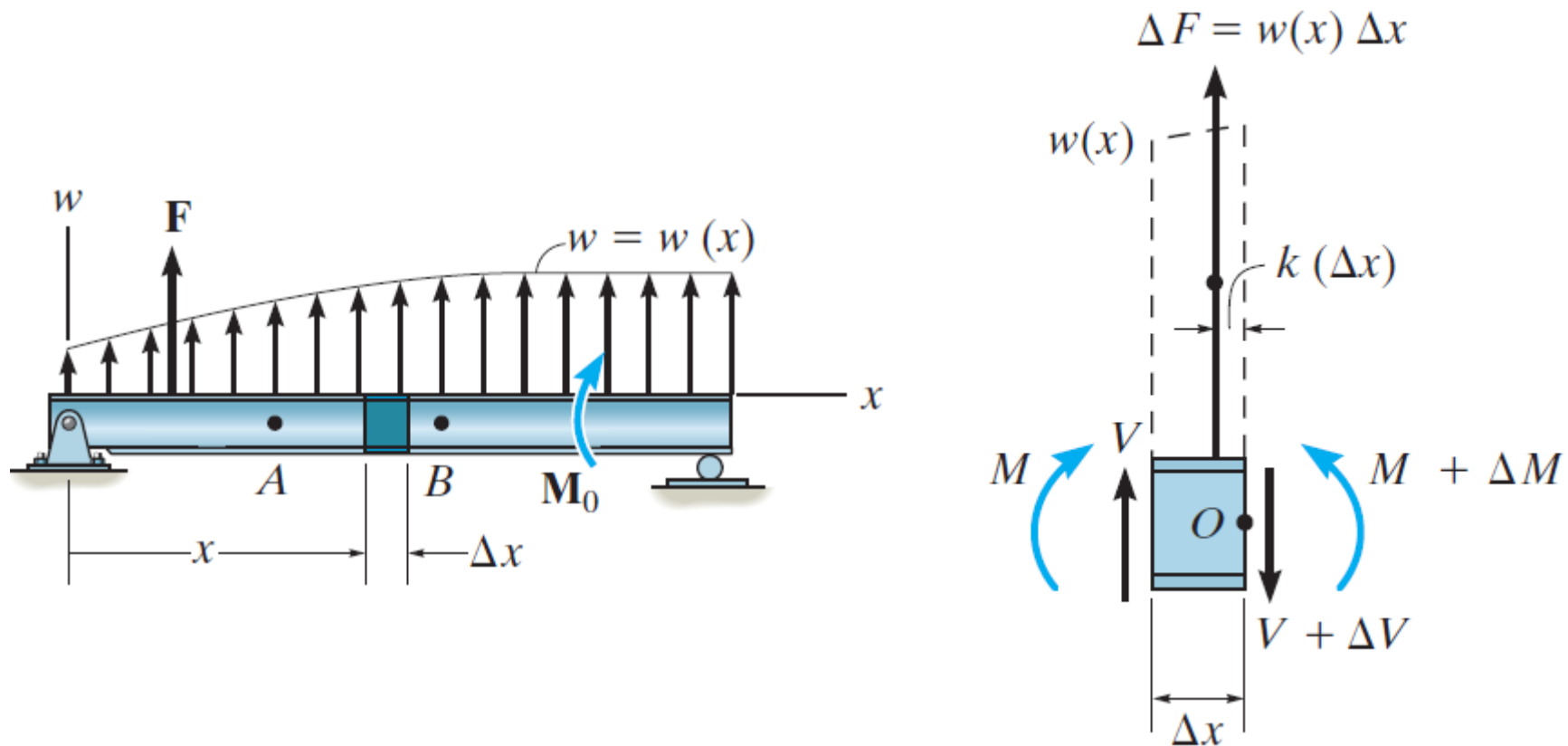


Example: Draw the shear and moment diagrams for the beam shown below.

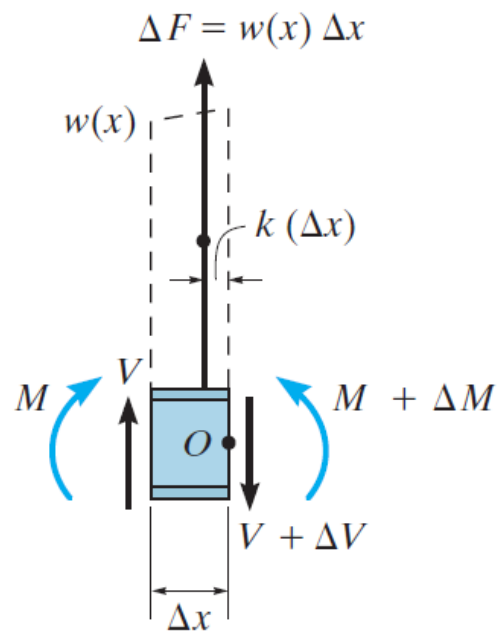


Relations Among Distributed Load, Shear, and Moment

Distributed Load



Relation between the Distributed Load and Shear



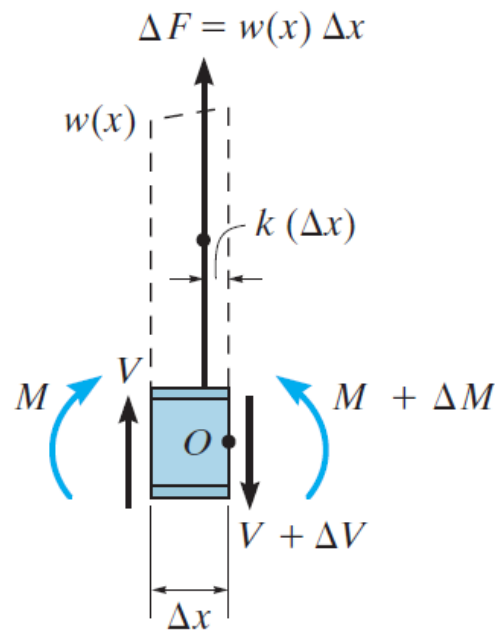
$$\frac{dV}{dx} = w(x)$$

Slope of
shear diagram = Distributed load
intensity

$$\Delta V = \int w(x) dx$$

Change in
shear = Area under
loading curve

Relation between the Shear and Moment



$$\frac{dM}{dx} = V$$

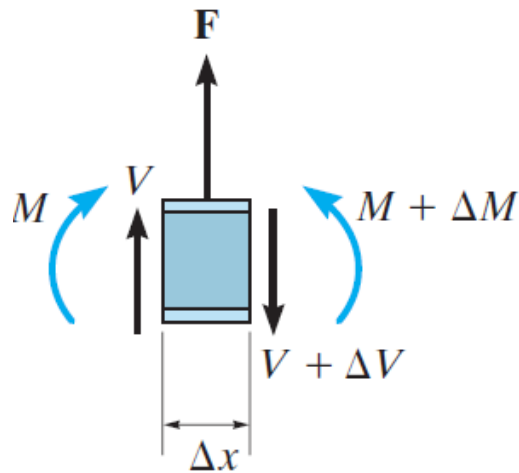
Slope of
moment diagram = Shear

$$\Delta M = \int V dx$$

Change in
moment = Area under
shear diagram

Notice that a maximum bending moment $|M|_{\max}$ will occur at the point where the slope $dM/dx = 0$, and this is where the shear is equal to zero.

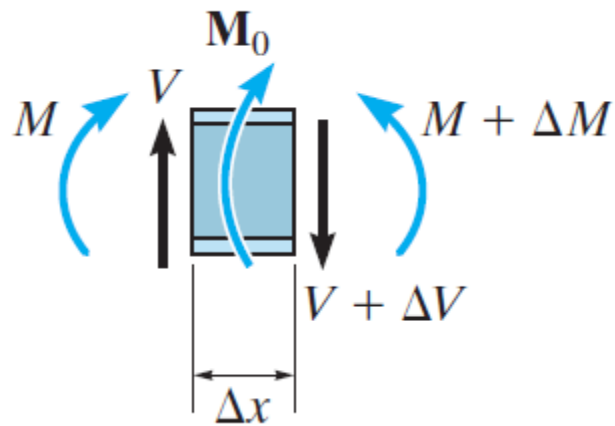
Force



$$\Delta V = F$$

Since the change in shear is positive, the shear diagram will “**jump**” upward when $\mathbf{F}$ acts upward on the beam.

Couple Moment



$$\Delta M = M_0$$

The change in moment is positive, or the moment diagram will “**jump**” upward if $\mathbf{M}_0$ is clockwise. Likewise, the jump $\Delta \mathbf{M}$ is downward when $\mathbf{M}_0$ is counterclockwise.



This concrete beam is used to support the deck. Its size and the placement of steel reinforcement within it can be determined once the shear and moment diagrams have been established.

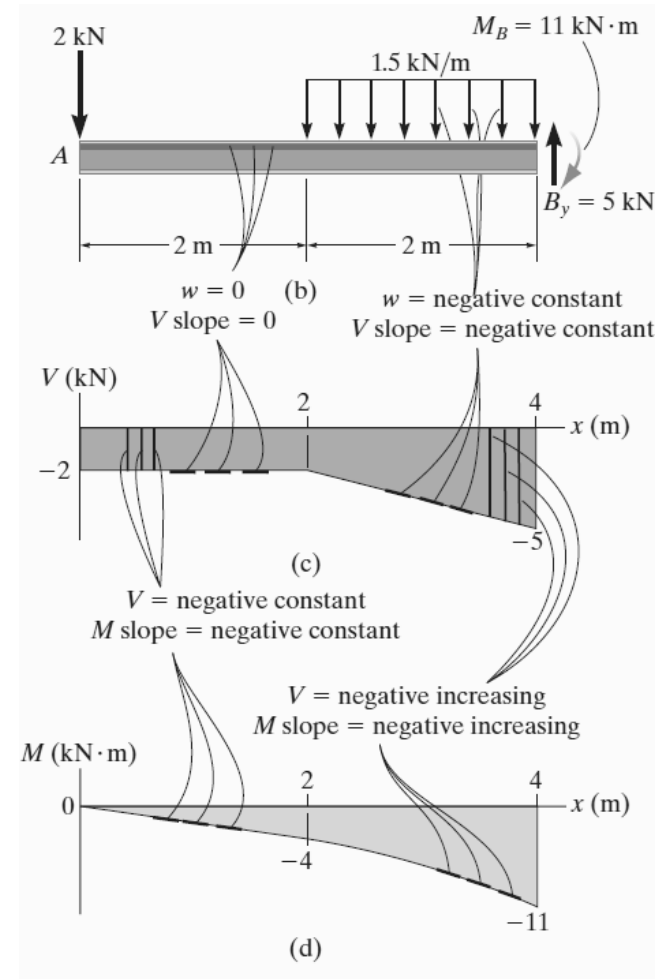
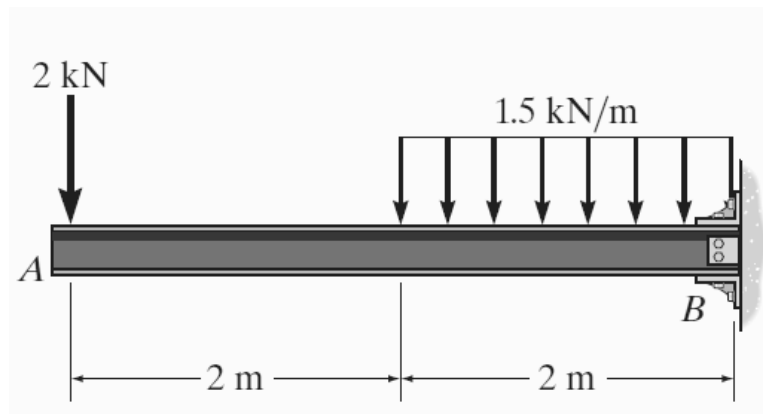
Concluding Notes

- The slope of the shear diagram at a point is equal to the intensity of the distributed loading, where positive distributed loading is upward, i.e., $dV/dx = w(x)$.
- The change in the shear ΔV between two points is equal to *the area* under the distributed-loading curve between the points.
- If a concentrated force acts upward on the beam, the shear will jump upward by the same amount.
- The slope of the moment diagram at a point is equal to the shear, i.e., $dM/dx = V$.
- The change in the moment ΔM between two points is equal to the *area* under the shear diagram between the two points.

Concluding Notes

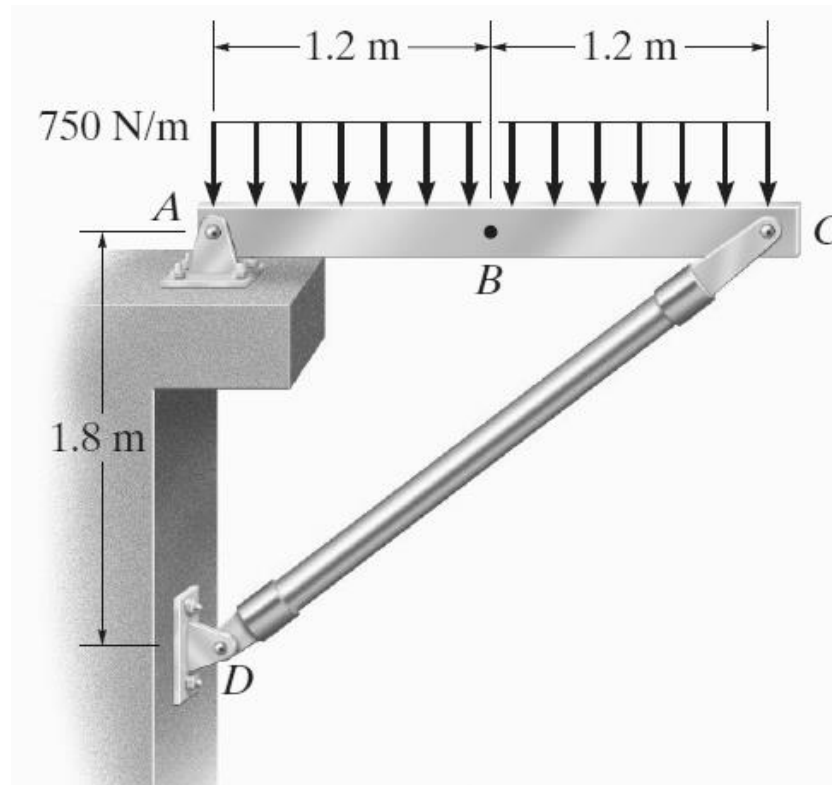
- If a *clockwise* couple moment acts on the beam, the shear will not be affected; however, the moment diagram will jump *upward* by the amount of the moment.
- Points of *zero shear* represent points of *maximum or minimum moment* since $dM/dx = 0$.
- Because two integrations of $w = w(x)$ are involved to first determine the change in shear, $\Delta V = \int w(x) dx$, then to determine the change in moment, $\Delta M = \int V dx$, then if the loading curve $w = w(x)$ is a polynomial of degree n , $V = V(x)$ will be a curve of degree $n + 1$, and $M = M(x)$ will be a curve of degree $n + 2$.

Example: Draw the shear and moment diagrams for the cantilever beam below.



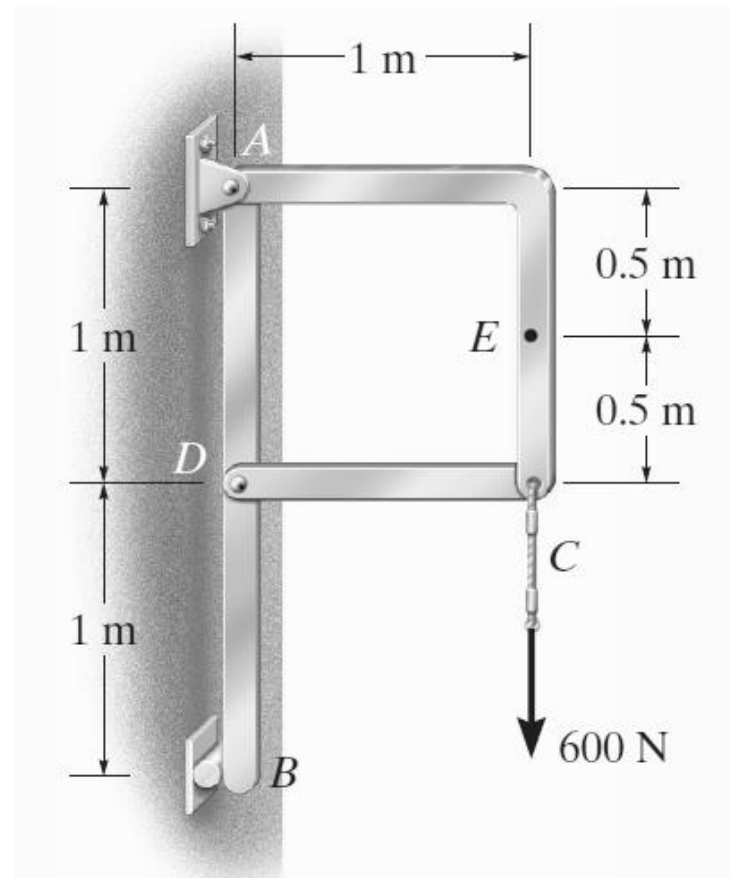
Practice at Home

1. Determine the normal force, shear force, and bending moment acting at point B of the two-member frame shown below.



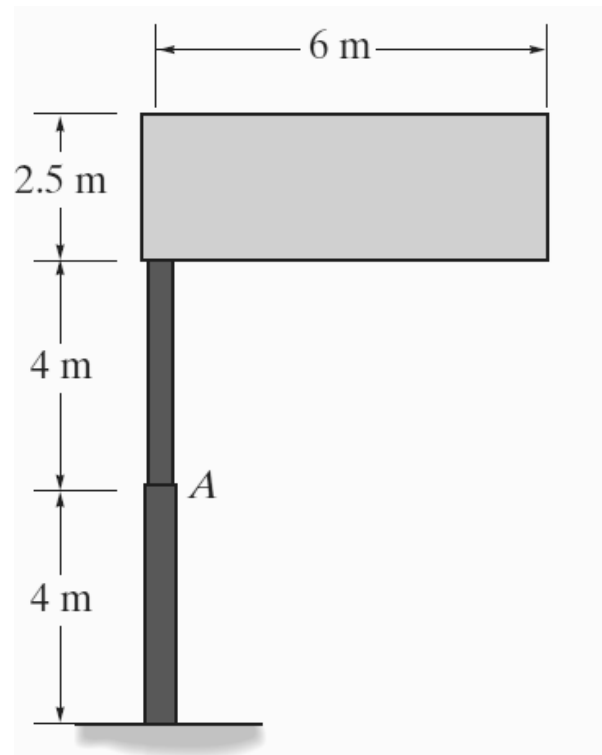
Practice at Home

2. Determine the normal force, shear force, and bending moment acting at point E of the frame loaded as shown below.



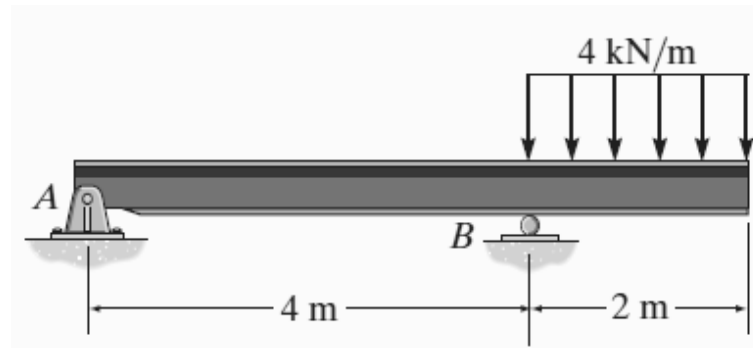
Practice at Home

3. The uniform sign shown below has a mass of 650 kg and is supported on the fixed column. Determine the internal loadings at A if the maximum uniform wind loading on the sign is 900 Pa.

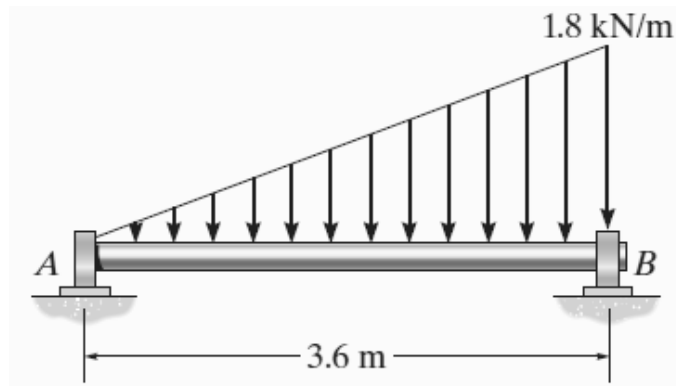


Practice at Home

4. Draw the shear and moment diagrams for the overhang beam below.

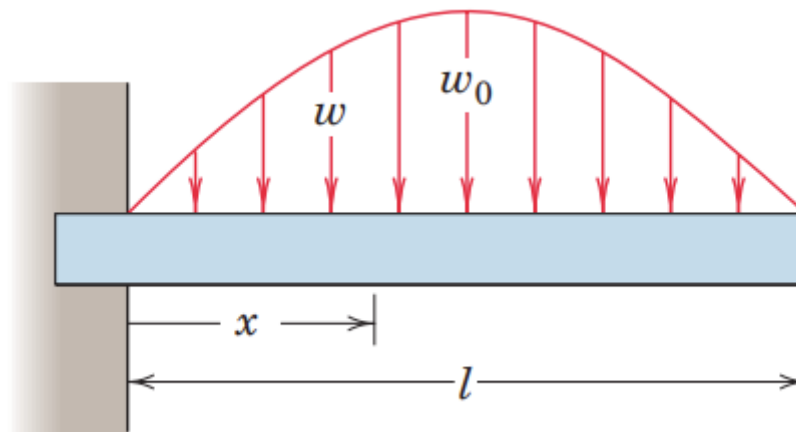


5. The shaft below is supported by a thrust bearing at *A* and a journal bearing at *B*. Draw the shear and moment diagrams.



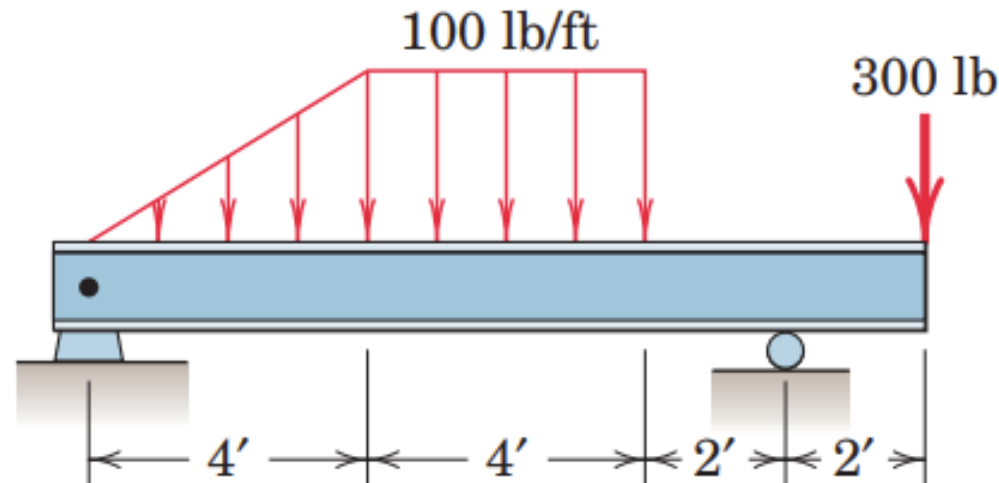
Practice at Home

6. The cantilever beam is subjected to the load intensity (force per unit length) which varies as $w(x) = w_0 \sin(\pi x/l)$. Determine the shear force V and bending moment M as functions of the ratio x/l .



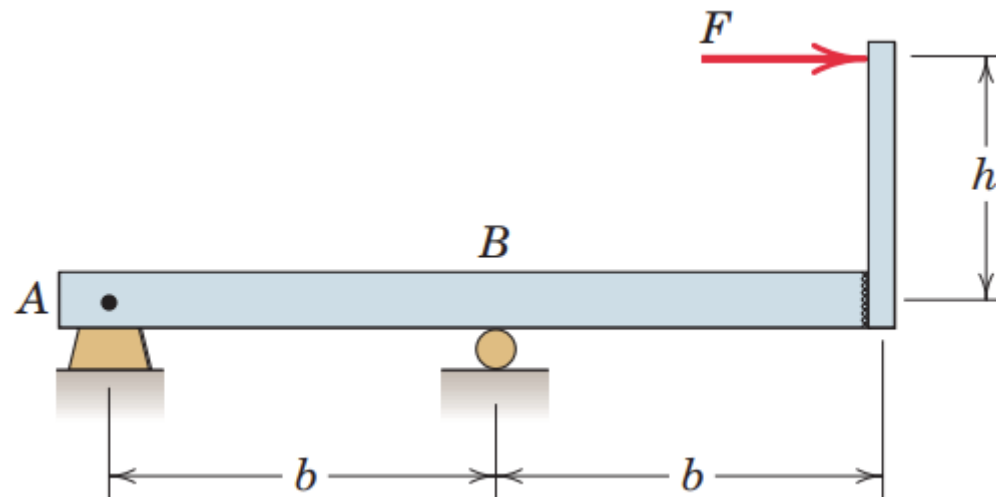
Practice at Home

7. Draw the shear-force and bending-moment diagrams for the loaded beam and determine the maximum moment M and its location x from the left end.



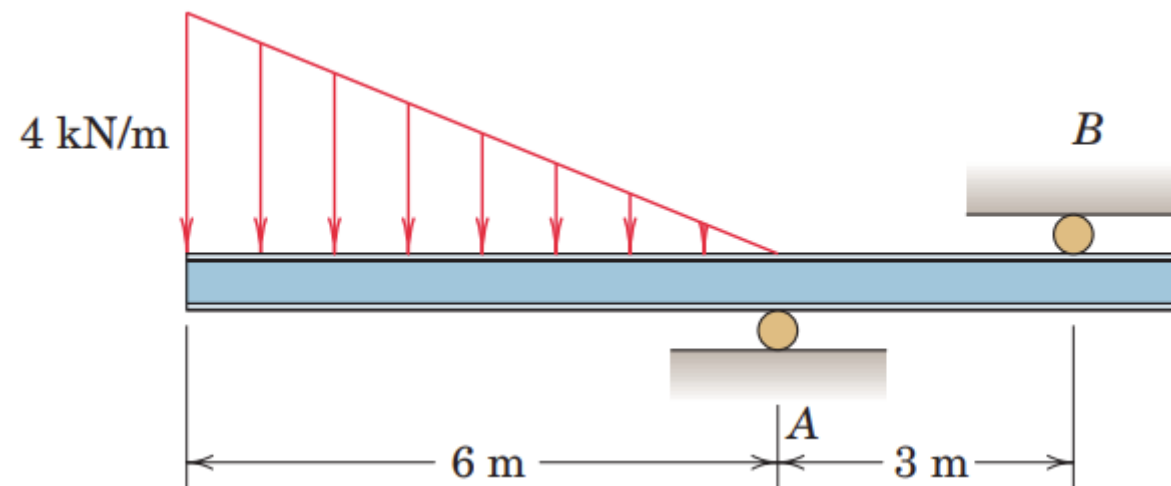
Practice at Home

8. Draw the shear and moment diagrams for the beam loaded by the force F applied to the strut welded to the beam as shown. Specify the bending moment at point B .



Practice at Home

9. Draw the shear and moment diagrams for the loaded beam and specify the shear V and moment M at a section 3 m to the left of the support at A .



Practice at Home

10. The beam supports a uniform unit load w . Determine the location x of the two supports so as to minimize the maximum bending moment $M_{\max}$ in the beam. Specify $M_{\max}$.

